DBMS – SQL QUERY IMPLEMENTATION FOR E-COMMERCE DATABASE

1.Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.

Display all products

```
SQL> SELECT * FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                         CATEGORY_ID
---------- -------------------------------------------------- ----------
       101
Women Kurti
       300 Women

       102
Men Shirt
      1299 Men

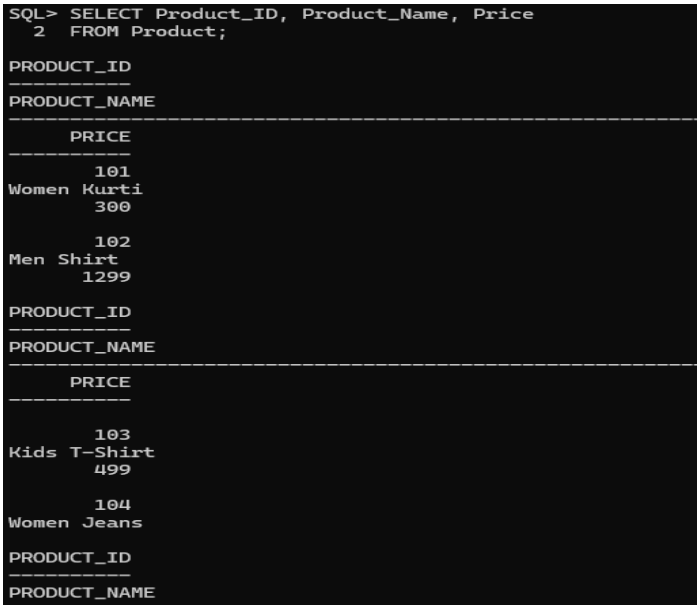
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                         CATEGORY_ID
---------- -------------------------------------------------- ----------

       103
Kids T-Shirt
       499 Kids

       104
Women Jeans

PRODUCT_ID
----------
PRODUCT_NAME
```

Display selected product Details:



```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------
       101
Women Kurti
       300

       102
Men Shirt
      1299

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------
     PRICE
----------

       103
Kids T-Shirt
       499

       104
Women Jeans

PRODUCT_ID
----------
PRODUCT_NAME
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition.

Find products with price less than 1000 Find products

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price < 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------
       101
Women Kurti
       300 Women

       103
Kids T-Shirt
       499 Kids

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------

       106
Cotton Bag
       300                                                            102

       107
Eco Bag
```

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 100;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------
       101
Women Kurti
       300 Women

       102
Men Shirt
      1299 Men

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------

       103
Kids T-Shirt
       499 Kids

       104
Women Jeans
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price ASC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- ---------------------------------------------------- -----------
       117
Paper Pen
        50 Stationery                                                   104

       113
Bamboo Toothbrush
       100 Personal Care                                                105

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- ---------------------------------------------------- -----------

       112
Eco Notebook
       150 Stationery                                                   104

       109
Paper Bags

PRODUCT_ID
----------
```

Products from highest to lowest price

```
SQL> SELECT *
  2  FROM Product
  3  ORDER BY Price DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- ---------------------------------------------------- -----------
       105
Sports Shoes
      2499 Footwear

       104
Women Jeans
      1599 Women

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- ---------------------------------------------------- -----------

       118
Bamboo Chair
      1500 Furniture                                                    108

       102
Men Shirt

PRODUCT_ID
----------
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

```
SQL> SELECT DISTINCT Category
  2  FROM Product;

CATEGORY
----------------------------------------
Women
Men
Kids
Footwear

Stationery
Personal Care
Kitchen
Home
Furniture

10 rows selected.
```

2. Search Products

2.1 Search Products Based on Price

This displays products whose price is between ₹100 and ₹400

```
SQL> SELECT Product_ID, Product_Name, Price
  2  FROM Product
  3  WHERE Price BETWEEN 100 AND 400;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
       101
Women Kurti
       300

       106
Cotton Bag
       300

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------

       107
Eco Bag
       250

       108
Bamboo Bottle
```

2.2 Search Products Based on Category

This displays products belonging to category 101.

```
SQL> SELECT Product_ID, Product_Name, Category_ID
  2  FROM Product
  3  WHERE Category_ID = 101;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
CATEGORY_ID
-----------
       107
Eco Bag
        101

       109
Paper Bags
        101

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------
CATEGORY_ID
-----------

       114
Jute Bag
        101
```

3. Retrieve Customer Information

3.1 Display all customers

```
SQL> SELECT * FROM Customer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
EMAIL
----------------------------------------
PHONE
--------------
ADDRESS
----------------------------------------
         1
Keerthi
keerthi@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
EMAIL
----------------------------------------
PHONE
--------------
ADDRESS
----------------------------------------
9876543210
Chennai
```

3.2 Display selected customer information

```
SQL> SELECT Customer_ID, Customer_Name, Email, Phone, Address
  2  FROM Customer;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------
          1
Keerthi
keerthi@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------
9876543210
Chennai
```

3.3 Search for a particular customer

```
SQL> SELECT *
  2  FROM Customer
  3  WHERE Customer_Name = 'Keerthi';

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------
           1
Keerthi
keerthi@gmail.com

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------
EMAIL
--------------------------------------------------------------------
PHONE
---------------
ADDRESS
--------------------------------------------------------------------
9876543210
Chennai
```

4. Apply Filtering Conditions

4.1 AND Condition

This displays products that satisfy both conditions.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price > 100
  4  AND Category_ID = 101;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                            CATEGORY_ID
---------- -------------------------------------------------- -----------
       107
Eco Bag
       250                                                           101

       109
Paper Bags
       150                                                           101

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                            CATEGORY_ID
---------- -------------------------------------------------- -----------

       114
Jute Bag
       250 Women                                                     101
```

4.2 OR Condition

This displays products belonging to either category 101 or 102.

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID = 101
  4  OR Category_ID = 102;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------
       106
Cotton Bag
       300                                                            102

       107
Eco Bag
       250                                                            101

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------

       108
Bamboo Bottle
       350                                                            102

       109
Paper Bags
```

4.3 IN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID IN (101, 102);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                            CATEGORY_ID
---------- -------------------------------------------------- -----------
       106
Cotton Bag
       300                                                            102

       107
Eco Bag
       250                                                            101

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                            CATEGORY_ID
---------- -------------------------------------------------- -----------

       108
Bamboo Bottle
       350                                                            102

       109
```

4.4 BETWEEN Operator

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Price BETWEEN 100 AND 1000;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------

       101
Women Kurti
       300 Women

       103
Kids T-Shirt
       499 Kids

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------

       106
Cotton Bag
       300                                                            102

       107
```

Additional NOT example:

```
SQL> SELECT *
  2  FROM Product
  3  WHERE Category_ID NOT IN (101, 102);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------
       110
Wooden Brush
       200                                                            103

       112
Eco Notebook
       150 Stationery                                                 104

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE CATEGORY                                             CATEGORY_ID
---------- -------------------------------------------------- -----------


       113
Bamboo Toothbrush
       100 Personal Care                                              105

       115
Steel Bottle
```

5. Basic Business Reports

5.1 Product Price Report



5.2 Customer Order Report

```
SQL> SELECT C.Customer_ID, C.Customer_Name,
  2         O.Order_ID
  3  FROM Customer C
  4  JOIN Orders O
  5  ON C.Customer_ID = O.Customer_ID;

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID
----------
         1
Keerthi
      1001

         2
Priya
      1002

CUSTOMER_ID
-----------
CUSTOMER_NAME
----------------------------------------
  ORDER_ID
----------

         3
Divya
      1003
```

5.3 Product Order Report

```
SQL> SELECT P.Product_ID, P.Product_Name,
  2         OD.Order_ID, OD.Quantity, OD.Subtot
  3  FROM Product P
  4  JOIN Order_Details OD
  5  ON P.Product_ID = OD.Product_ID;

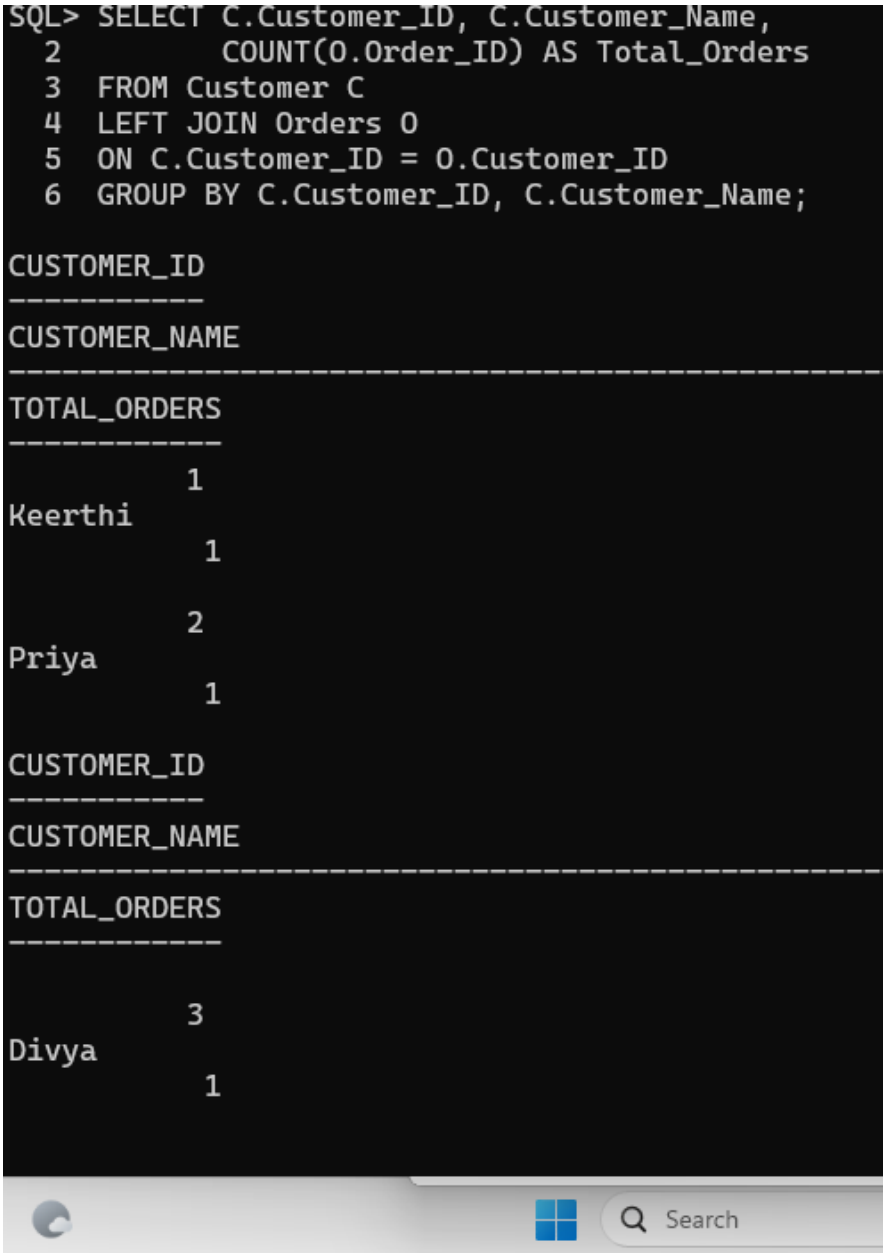
PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY   SUBTOTAL
---------- ---------- ----------
       101
Women Kurti
      1001          2       3000

       102
Men Shirt
      1001          1       2000

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------
  ORDER_ID   QUANTITY   SUBTOTAL
---------- ---------- ----------

       103
Kids T-Shirt
      1002          2       2000
```

5.4 Customer-wise Total Orders



5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT C.Customer_ID, C.Customer_Name,
  2         NVL(SUM(OD.Subtotal), 0) AS Total_Purchas
  3  FROM Customer C
  4  LEFT JOIN Orders O
  5  ON C.Customer_ID = O.Customer_ID
  6  LEFT JOIN Order_Details OD
  7  ON O.Order_ID = OD.Order_ID
  8  GROUP BY C.Customer_ID, C.Customer_Name;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------
TOTAL_PURCHASE
--------------
         1
Keerthi
         5000

         2
Priya
         4500

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------
TOTAL_PURCHASE
--------------

         3
Divya
```